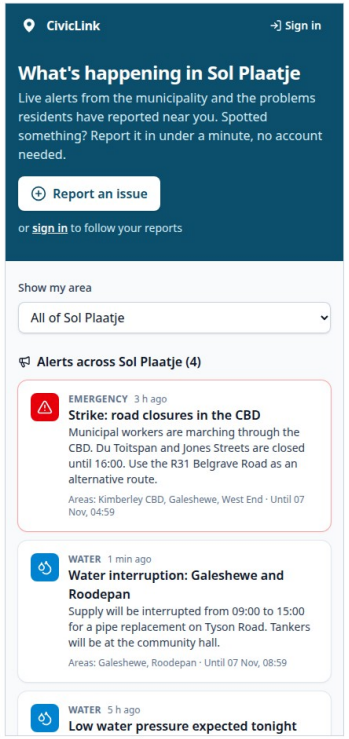
SOL PLAATJE LOCAL MUNICIPALITY · KIMBERLEY

CivicLink: reporting service problems and warning residents

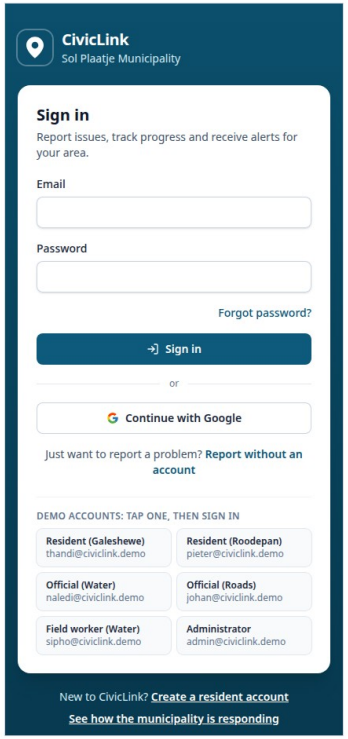
The problem Reports come in by phone, walk-in and WhatsApp, with no location or photo. The same burst pipe is reported many times, teams are dispatched by phone with no record, and residents never hear back.

31 March 2026 A workers' strike closed roads across Kimberley and there was no official way to warn people.

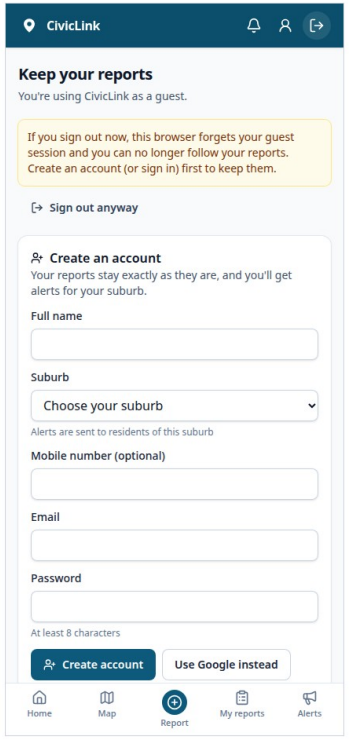
What I built One web app where a report goes from the resident's phone to the right department, to a field worker and back, with alerts and a full audit trail.



(a) Public home (visitor)



(b) Sign in



(c) Guest: keep reports

All 39 requirements built and tested (185 automated tests)

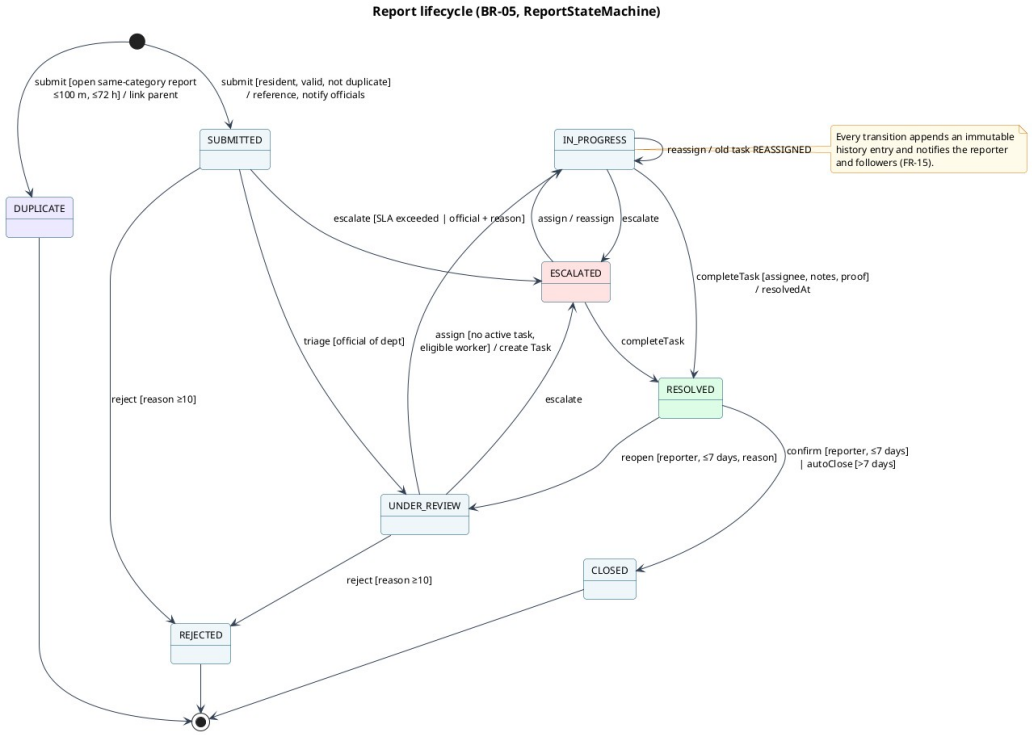
Residents report in 3 steps (map pin, photos, severity), even offline or as a guest; follow it live; confirm with a rating or reopen

Officials department queue, triage, assign one or many, escalate; alerts now or scheduled; analytics

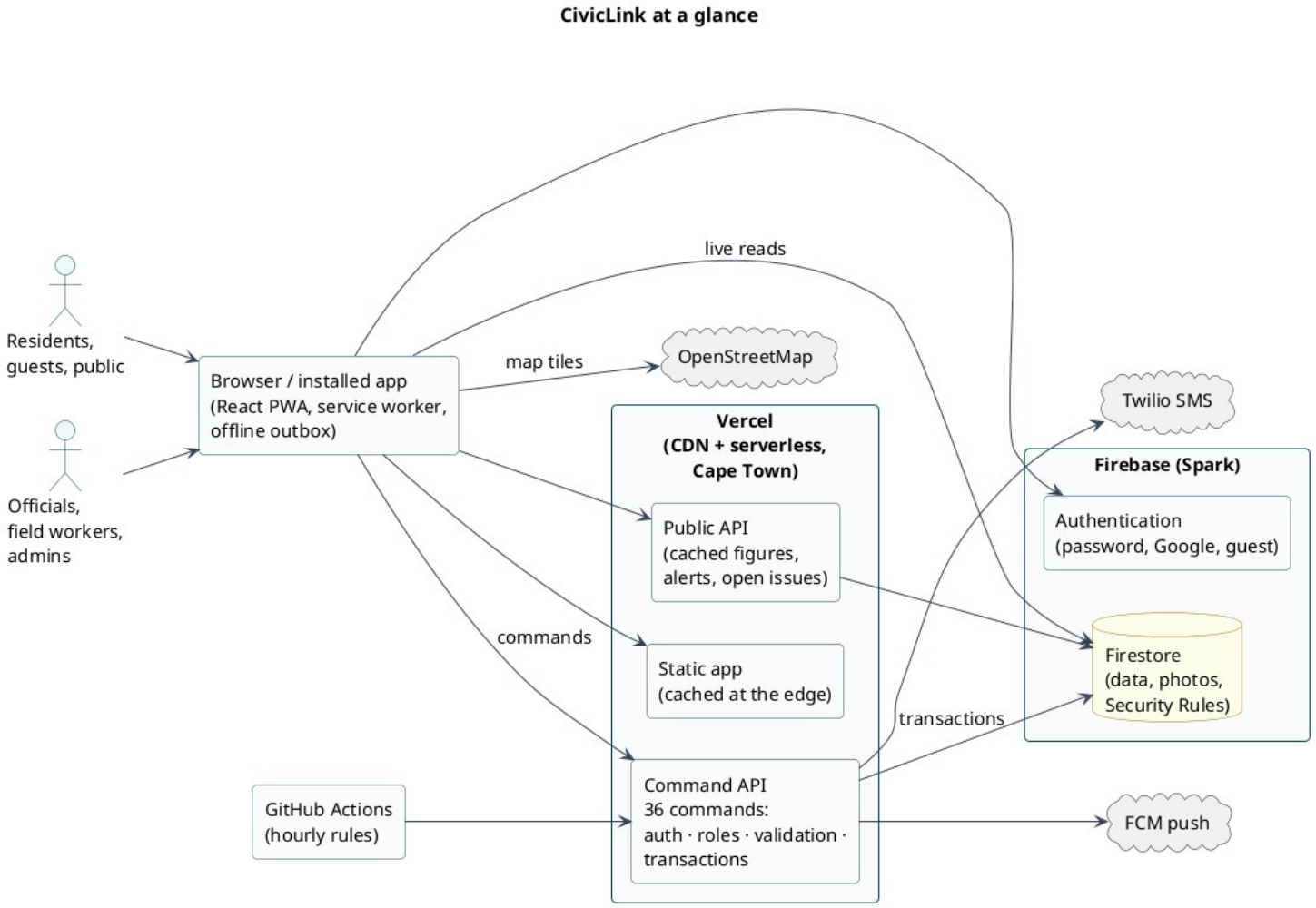
Field workers accept, start and complete tasks with notes and a proof photo

Administrators staff, roles, departments, SLA targets, system log, SMS outbox

Everyone public home with alerts and open issues nearby, transparency page, install on the phone



Reads from the database, changes through commands



Reads the app listens to Firestore directly, so screens update by themselves. Security Rules decide who can read what.

Changes 36 server commands: check the caller and role, validate with the same Zod schemas as the forms, then run a transaction.

Why Security Rules cannot check rules that span documents (reference numbers, duplicates, one active task).

Transactions and idempotent retries

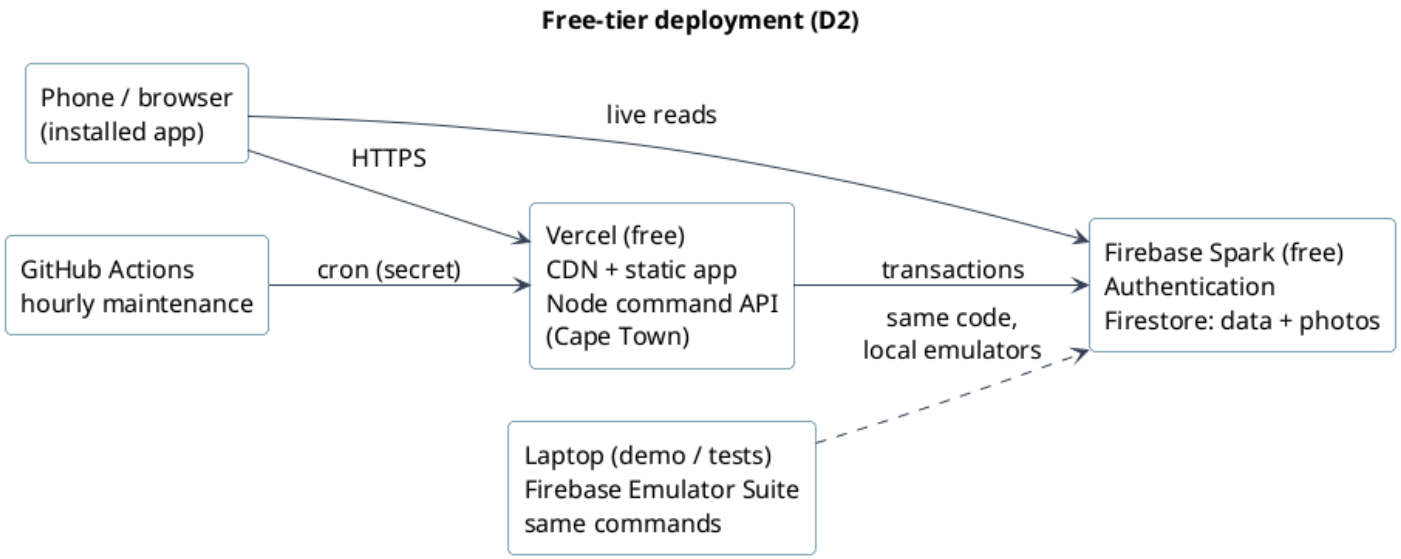
TX-1 submit takes the next reference number, searches for duplicates within 100 m and saves the report, history, map entry and notifications together.

TX-2 assign re-checks "one active task per report" inside the transaction: if two officials assign at the same moment, exactly one wins (test IT-ASG-07).

TX-3 complete the task and the report are resolved together or not at all.

Idempotent submit the phone sends a random clientId; the server builds the report id from the user and that clientId. If the same report arrives twice, it returns the first one instead of creating a duplicate (IT-OFF-01).

Running on free tiers without changing the code



Problem Cloud Functions and Cloud Storage need Firebase's paid plan.

Solution the same command registry runs on Vercel through a small Node adapter; Firebase stays on the free Spark plan.

Photos compressed in the browser to ≤ 700 KB and stored in Firestore; the server checks the real file type from its first bytes.

Schedules GitHub Actions calls the maintenance endpoint hourly (Vercel's free cron is daily).

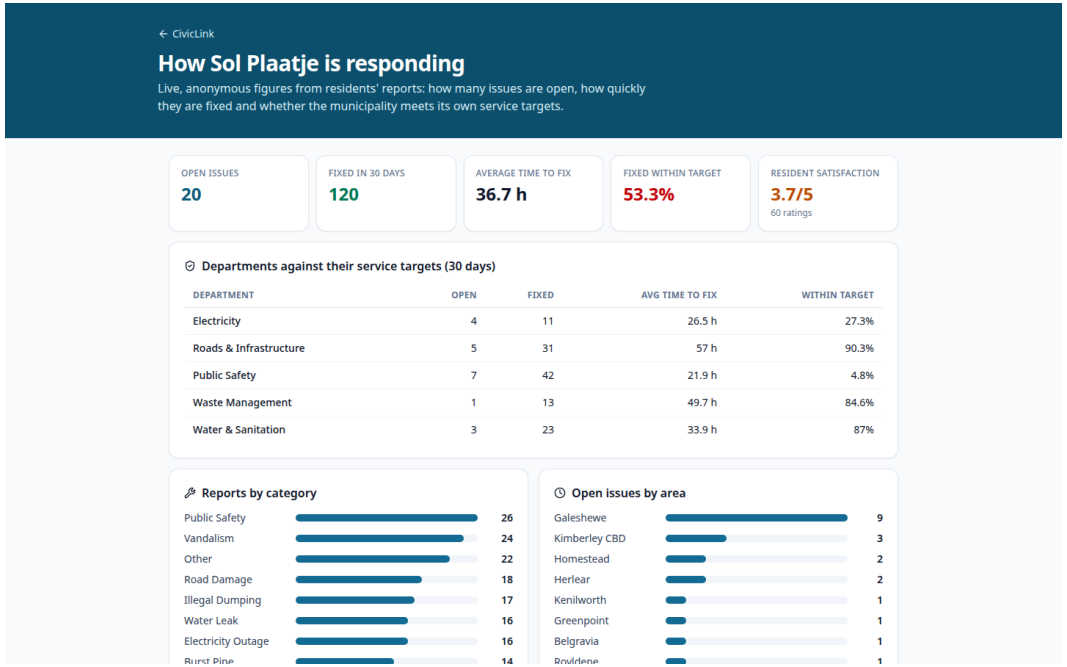
Caching and the CDN: fast for residents, cheap for the database

App files hashed build files are cached for a year ("immutable"); the page itself is checked on every visit, so updates appear immediately.

Public data the transparency figures (5 min) and the public feed (1 min) are cached at Vercel's edge. However many people visit, Firestore is read about once per window.

Server cache each server instance also keeps the last result for up to a minute.

Evidence NFT-11 on the live site: x-vercel-cache = HIT on repeat requests.



Installable app that keeps working offline

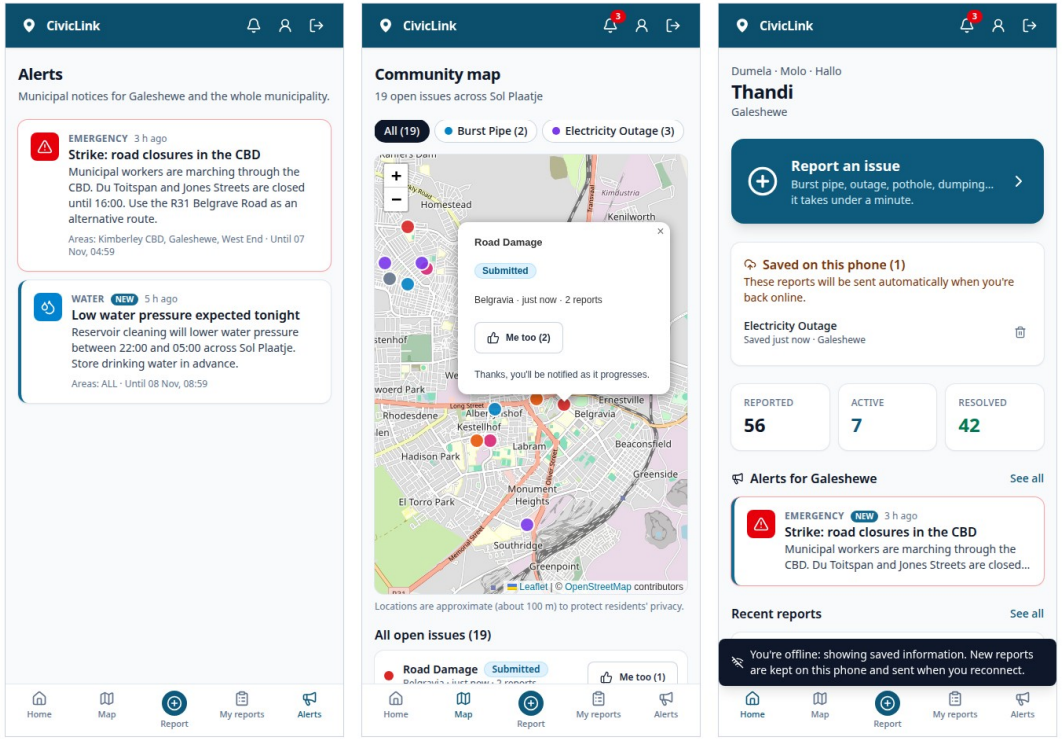
Install web app manifest + service worker: CivicLink installs on the home screen like an app.

Open offline at install the service worker saves every app file (the build writes the list into sw.js); pages fall back to the saved copy.

Saved data Firestore's offline cache shows reports and alerts already seen; up to 400 map tiles are kept.

Report offline "Save and send later" puts the report in an IndexedDB outbox; it is sent automatically on reconnect, with the same clientId.

Lessons the security policy also applies inside the service worker (DEF-14), and saved files must be matched ignoring Vary (DEF-16).



(a) Suburb alerts (b) "Me too" on the map (c) Offline: report saved

Maps and location without paid APIs

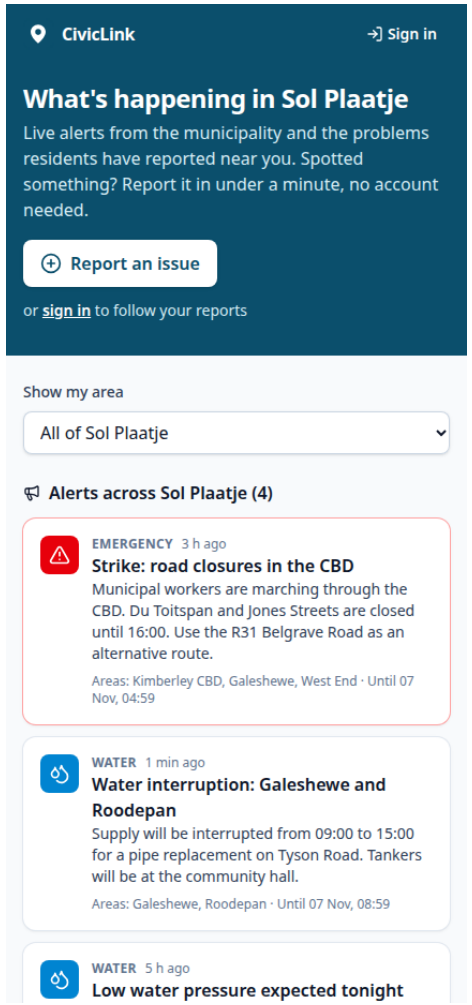
Leaflet + OpenStreetMap no API key or billing; the suburb comes from a list, so no reverse geocoding is needed.

Location GPS or tap/drag the pin; a position outside Sol Plaatje is explained instead of silently used, and the nearest suburb is suggested.

Duplicates each report stores a geohash; a range query finds open reports of the same category within 100 m and 72 h.

Privacy the public map uses locations rounded to about 100 m and never shows names, descriptions or photos.

Phones open issues are also listed under the map, with "Me too" to confirm an issue instead of reporting it again.



Reaching residents: in-app, push and SMS

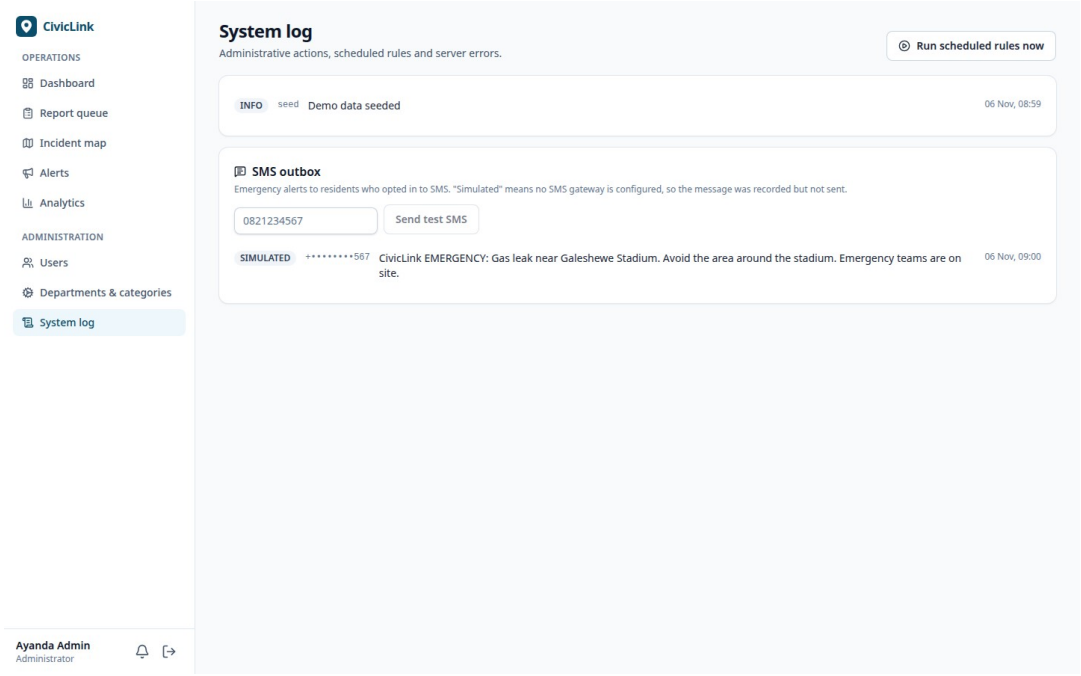
In-app notifications are written in the same transaction as the status change and appear live.

Web push messages are collected while a command runs and sent only after it commits, so a failed command never notifies anyone; dead device tokens are removed.

SMS Emergency alerts go by SMS to residents who opted in. An SmsGateway interface has a Twilio and a simulated implementation; every message is stored with its status.

Twilio trial trial accounts only send preset templates, so a setting sends the template while the real text stays in the outbox.

Scheduled alerts published by the maintenance run, claimed in a transaction so they go out exactly once.



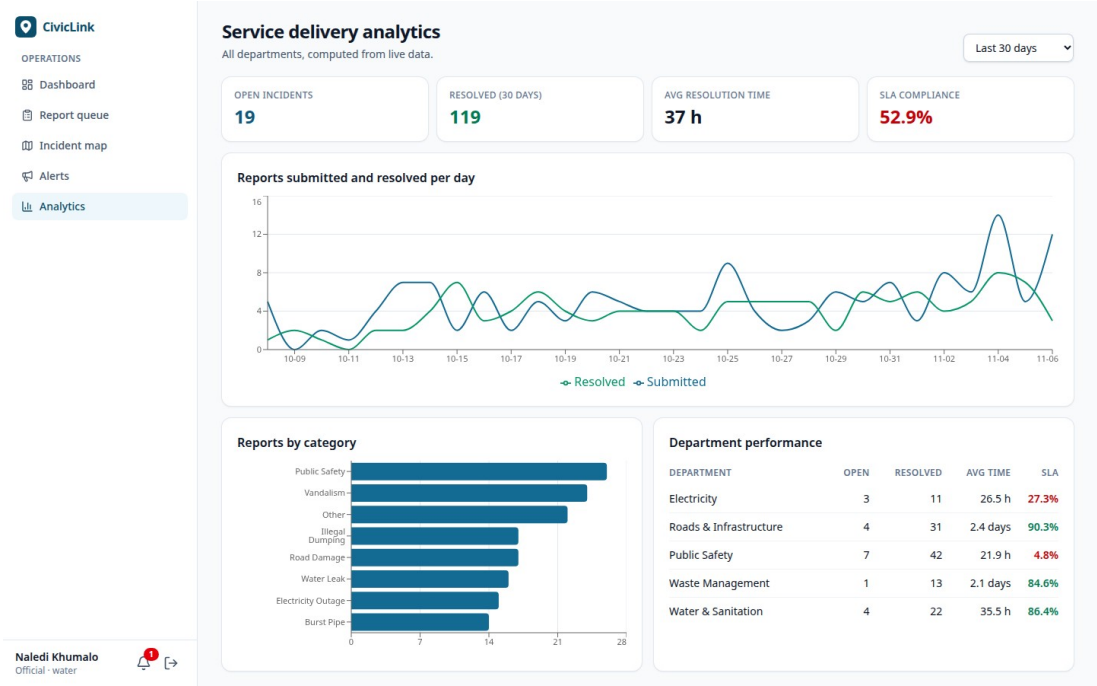
Analytics computed from live data

How the server reads the reports and runs one pure function, computeAnalytics, that is unit-tested against a known dataset.

Figures open reports, resolution time, SLA compliance per department, 30-day trend and the average resident rating.

Empty data returns "no data" instead of NaN (UT-ANL-01).

Public version the same function feeds the transparency page, but only counts and averages leave the server.



Security and privacy

Roles signed custom claims + Security Rules for every read; every command re-checks the role and refuses deactivated accounts.

Validation the same Zod schemas in the forms and on the server; photo types checked from their bytes.

Browser enforced Content Security Policy and HSTS; no eval, so the validation library runs in "jitless" mode.

Guests (D4) anonymous Firebase sign-in, so all the normal rules still apply; 3 reports an hour, 10 guest sessions per network per hour.

POPIA residents can delete their account; reports stay as "Former resident"; data stays in South Africa.

185 tests passing, all five success criteria met

Levels unit (41), security rules (22), API integration (99), end-to-end/PWA/live site (23); CI on every push.

Performance update reaches the resident in 1.9 s (target 5 s); screens < 1.4 s; p95 command 73 ms; 0 accessibility and 0 CSP violations.

Live checks the same tests run against the deployed site; push and SMS checked on a real phone.

Defects 17 logged and fixed, several found only on the live site or a phone (tiles, offline, notifications panel).

Next paid SMS gateway with a queue, a paid scheduler, and a pilot with one department.